A Non-Intrusive Multi-Modal Steering-Wheel Interface with Edge-Embedded Sensor Fusion and an AI Vision Fast-Path Interlock for Physiological Driver Drowsiness Detection

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Abstract

Driver fatigue is a major cause of road accidents, but commercial countermeasures use camera-based methods to monitor eye-blinking and head pose, which are vulnerable to occlusion, poor lighting, and privacy concerns. In this paper, we introduce a passive and multimodal steering-wheel interface that attacks the autonomic nervous system by means of the driver's hands. Galvanic Skin Response (GSR) and single-lead Electrocardiogram (ECG) sensors placed in the rim of the wheel collect GSR and SDNN, while an IMU filters artifacts from the steering motion. An auxiliary AI Vision Layer runs Media-Pipe Face Mesh on a laptop, detects Eye Aspect Ratio (EAR), and sends EYE CLOSED and FATIGUE binary flags to the microcontroller over USB-serial connection. An interlock fusion rule combines an immediate vision lock-out and a cross-validated physiological fallback aimed to prevent false alarms, for which there has been no test under false-positive inducing conditions yet. We have conducted five bench-test stages using a breadboard implementation. Embedded R-peak detection has shown more than 90% informal visual concordance with the reference waveform display, SDNN and resting GSR values have been measured within the known ranges, the fused alert logic worked reliably in small-scale repeats (n=3-5). The average latency of web-camera to microcontroller. Webcam-to-microcontroller transmission latency was about 100 ms; the end-to-end alert latency, including the 333 ms EYE_CLOSED confirmation window, was about 333-400 ms. Since the physiological protocol triggered seated relaxation in awake volunteers without inducing drowsiness, the SDNN-rise/GSR-slope-decline feature set is validated only as a relaxation-induced autonomic change; use as a drowsiness precursor is a hypothesis to be tested using a driving simulator. System technology readiness level is 3, which proves that a camera-free, privacy-respecting, edge-native architecture can be built on commodity hardware.

Keywords: driver drowsiness detection; edge computing; galvanic skin response; heart rate variability; sensor fusion; ESP32 microcontroller; Media-Pipe Face Mesh; advanced driver-assistance systems

1. Introduction

Road-traffic injuries are among the most persistent global health problems, and among their causes, driver drowsiness is the one that holds the most dangerous ground since it impairs drivers' ability to recognize potential dangers. According to U.S. federal crash statistics, there were 693 fatalities due to

drowsy-driver crashes in 2022, which is considered a substantial underreporting [1]. This problem has been resisting two decades of ADAS deployment in part due to the lack of physiological fatigue detection.

In contrast to the clear-cut condition of alcohol intoxication, drowsiness is a complex, gradual neurophysiological state. Because of the brain's amazing ability to shield initial signs of cognitive tiredness from conscious realization, drivers are poor evaluators of their own state and seldom realize that they have reached microsleep stage when they are no longer able to control the car. The above problem justifies the need for an objective monitoring system.

The most common paradigm in commercial driver-monitoring systems is the visual one, which means that a cabin-mounted camera is looking at PERCLOS (percentage of eyelid closure), yawn rate, and head posture [2], [3]. Even though such a paradigm is highly efficient in detection of fatigued drivers, it is inherently reactive: visual manifestations of the impairment happen no more than one to three minutes prior to crash occurrence.

Camera-based solutions also face a certain number of practical issues that have been described in literature before: sensitivity to ambient cabin light, obstruction from sunglasses, facial hair, or medical masks, lower accuracy when measuring drivers from different demographic backgrounds, and, last but not least, growing consumer concern about privacy in relation to video recording in the cabin [2], [4]. This problem has led to research into alternative methods of measuring driver's state, which would allow observing changes in the autonomic nervous system (ANS), which always precedes fatigue-related changes in behaviour by a much greater period of time. This is the hypothesis tested in this paper's bench validation (Sections 4 and 5).

To date, physiological approaches have, however, concentrated largely on wrist bands, ear-based sensor, and chest strap devices that monitor electrodermal activity, photoplethysmography, or electrocardiograms [5]–[8]. These solutions do provide true predictive advance warning capability, but the requirement to charge and pair the device before each use is an enormous barrier that prevents adoption by fleets or consumers and is often mentioned as the single biggest issue holding back physiological drowsiness detectors from deployment on production vehicles.

Yet another limiting factor present in almost all physiological systems described in literature is the way the architecture is designed: signal processing is often handled by a mobile app or backend system [9], [10]. Not only does this approach introduce extra latency and potential connection failure due to low coverage but it also implies sending biometric data out of the car, which creates another privacy hazard.

As a result, regulatory momentum has made the deployment of a driver state monitoring system an immediate necessity, not an optional comfort system. The EU's General Safety Regulation now mandates a Driver Drowsiness and Attention Warning (DDAW) system on all vehicle type approvals, with similar requirements being considered in other key markets [11]. As such, there is additional motivation to resolve the camera-based versus intrusive wearable problem outlined above with an approach that is at once non-invasive, predictive, and self-sufficient.

This research is inspired by the idea that the steering wheel represents an under sampled sensing area. Given that a driver maintains a near-constant contact with the wheel rim throughout the vast majority of his/her trip, dry-contact electrodes can in principle record both the GSR and single-lead ECG data without the need for any additional peripheral equipment to be worn or attached to the driver, which has not been extensively explored in the available literature.

However, extracting biomarkers associated with drowsiness from data recorded by steering wheel-mounted electrodes represents its own engineering challenge: Artifacts caused by normal steering actions affect both the GSR and ECG streams, and the processing pipeline needs to operate under the memory and energy constraints of an automotive-grade embedded microcontroller and not that of a smartphone or cloud-based GPU. The approach should be that of a lightweight and deterministic signal processing pipeline with a motion artifact rejection process, as opposed to the deep learning pipelines used in visual only prior art [3].

The paper is able to address the above challenge directly by presenting and validating on bench level a solution based on the fusion of sympathetic arousal extracted from the GSR signal, HRV extracted from the ECG signal, and a motion rejection mechanism using the IMUs in a determinist autonomic-shift finite state machine, and adding as a secondary mechanism the use of a vision-based layer based on the same video captured by the camera in a real time safety interlock layer without replacing the physiological solution as the main detection mechanism.

1.1 Research Objectives

Objective 1: Design a non-invasive, dry-contact sensing interface for integrating into the steering wheel of a vehicle capable of real-time acquisition of GSR and ECG signals without any need for wearables or user cooperation.

Objective 2: Design a deterministic, edge-native processing pipeline for fusion of GSR slope, ECG SDNN, and IMU-based motion-artefact rejection, which operates within the constraints of a standard microcontroller without needing offloading to cloud or smartphone**.**

Objective 3: Design a complementary AI Vision Layer based on Media-Pipe Face Mesh and the Eye Aspect Ratio (EAR), designed as a rapid behavioural interlock fused with the physiological pipeline via a cross-validated decision rule meant to reduce false positives.

Objective 4: Build a hardware prototype and validate its performance in terms of signal acquisition, correctness of the algorithms, and fused alerts, thereby determining the Technology Readiness Level of the system and setting up future validation goals.

2. Related Work

Research in automated driver state monitoring can be divided into three major categories based on methodology; camera-based behaviour analysis, physiologically wearable devices, and the third category which is the smallest but newest of all, which involves non-wearable physiology-based sensors integrated

with the existing contact points inside the vehicle. The current review will contextualize the proposed approach in the context of steering wheel in contrast with all three categories.

Camera-based systems have been implemented by Nath et al. [2], which use Media-Pipe Face Mesh and eye aspect ratio to detect drowsiness in real-time using only single-board computers for software-based alerts without need for a dedicated sensor. Although this approach is computationally cheap, it is limited in that it does not fuse any physiological biomarkers, does not provide any cross validation using autonomic responses, and is also affected by occlusion.

Ghoddoosian et al. [3] have provided a soft-label dataset and deep learning framework for EAR-based drowsiness detection using facial landmark regression from video, enhancing the classification robustness over the hard-threshold approach for EAR calculation. The work, however, continues to be dataset dependent, requires non-negligible inference cost in relation to microcontroller-class hardware, and, crucially, for the predictive onset case argued in Section 1, does not provide a means of detecting the pre-behavioural autonomic changes prior to eyelid closure.

Uccello et al. [4] performed a review of driver state analysis by means of hybrid measurements and machine learning, and the findings are directly pertinent: the review specifically notes the industry focus on reactive, camera, and vehicle dynamics (e.g., lane departure) monitoring techniques and the untapped potential of sensor fusion including physiological sensors, which is precisely what this paper seeks to achieve.

Regarding wearable physiological approaches, Amidei et al. [8] demonstrated GSR signals obtained from an Empatica E4 wristband using ensemble machine learning classifiers, showing the discriminatory power of GSR for drowsiness but only validating it on a wearable device that necessitates active cooperation and is limited in terms of battery capacity the steering wheel solution is specifically designed to eliminate.

Mai et al. [7] took the physiological approach centred around the ear region using physics-informed neural networks for edge inference but, similar to [8], are inherently tied to a wearable accessory which, in turn, is ruled out in advance in the current solution due to its non-wearable nature.

AlArnaout et al. [10] proposed HRV-based drowsiness detection using PPG sensors and machine learning, however, their solution transmits raw sensor data to a smartphone and, further, to a cloud server for inference introducing the latency and connectivity issues that are avoided in the edge native design.

The smallest and most architecturally significant family of physiological sensors integrated into the steering wheel itself is presented in the work by Babusiak et al. [5], who used a combination of ECG, PPG, and IMU sensors on an ATSAML21J18B microcontroller in a smart-steering-wheel prototype, and the work by Abdul Razak et al. [9], who used a steering wheel equipped with an AD8232 ECG module and HRV calculation based on Arduino. Both works confirm the potential of the steering wheel as a sensing interface, but Babusiak et al. note very high sensitivity to motion artefacts without a localised predictive fusion algorithm, while Abdul Razak et al. aim at monitoring of the cardiac health rather than drowsiness onset and use an external Wi-Fi module rather than purely edge decision making, – two problems that the proposed IMU-gated deterministic fusion pipeline directly solves.

The work by Essahraui et al. [6] introduced FastKAN-DDD, a Kolmogorov–Arnold Network architecture tailored specifically for TinyML application, showing that lightweight neural architectures are capable of delivering competitive drowsiness-classification accuracy even in a highly constrained setting. The method, however, was evaluated on primarily image-based datasets (e.g., UTA-RLDD) and not adjusted to the physiological tactile modality used in this work.

Table 1. Representative prior works and their relationship to the proposed system

Reference	Modality / Core Technology	Key Gap vs. Proposed System
Babusiak et al. [5], 2026	Steering-wheel ECG+PPG+IMU on ATSAML21J18B	High motion-artefact susceptibility; no localised predictive fusion algorithm
Essahraui et al. [6], 2025	FastKAN-DDD (TinyML) on image datasets	Not adapted to tactile steering-wheel physiological data
Mai et al. [7], 2025	Ear-worn physics-informed edge NN	Wearable-dependent; violates non-intrusive design requirement
Uccello et al. [4], 2025	Review of hybrid driver-state ML	Confirms industry reliance on reactive camera / kinematic systems
Abdul Razak et al. [9], 2024	Steering-wheel AD8232 ECG + Arduino HRV	Cardiac-health focus, not drowsiness; external Wi-Fi, not edge-only
AlArnaout et al. [10], 2025	PPG-based HRV via ML	Smartphone/cloud transmission introduces latency & dropout risk
Amidei et al. [8], 2023	GSR + ensemble ML on Empatica E4	Wearable wrist device; compliance burden; battery-limited
Nath et al. [2], 2024	MediaPipe/EAR on embedded SBC	Vision-only; no physiological fusion; reactive by nature
Ghoddoosian et al. [3], 2019	Deep-learning EAR regression + dataset	Dataset-dependent, high inference overhead, no physiological fusion

Gaps

Taken collectively, the surveyed literature shows that there is a clear structural gap in the current literature where no existing solution integrates dry-contact physiological sensing (GSR and ECG/HRV), sensor fusion algorithm, IMU-driven motion artefact rejection, and an AI-based vision cross-validated interlock within a single, non-wearable, off-cloud architecture deployed entirely in the steering wheel and column. Vision-only approaches are reactive and have privacy concerns; physiological wearables have predictive capabilities but come with compliance requirements; and although there have been some physiological prototypes that integrate with the steering wheel, none has had an algorithmic validation of their fusion nor has the processing been entirely edge-resident.

Problem Statement

There is a need for a driver-drowsiness detection solution that (i) obtains physiological biomarkers using a non-invasive method via a touchpoint already interacted with by the driver by default, (ii) conducts all critical inference processing on the edge to ensure low latency and privacy of the biometric data, (iii) is resilient against motion artefacts resulting from natural steering actions, and (iv) still has the fast, vision-derived interlock mechanism to act as a redundant safety feature without the false-positive and privacy issues associated with vision-only approaches. This paper tackles the stated problem.

3. Materials and Methods

Here, we detail the physical configuration of the sensor network and embedded hardware, the mathematically formulated edge-based processing pipeline for the signals, the fusion scheme along with the motion artefact rejection process and also that of the companion AI-based Vision Layer. There is no preexisting dataset to train the models, and instead, the system uses threshold-based detection based on physiology, which has been experimentally verified using a bench test setup explained in Section 4.

3.1 Physical Arrangement and Hardware Interface

The steering wheel is fitted with electrode pairs in the area where one would be gripping the steering wheel (at 10 o'clock and 2 o'clock points). There is an auxiliary set of conductive plates reaching out to 9 and 3 o'clock to ensure that contact will be achieved irrespective of the gripping positions. In contrast to the conventional system of disposable Ag/AgCl electrodes and gel that is deployed in the clinical environment, here the specifications involve robust metallic conductors that can withstand mechanical wear. These are matte polish stainless steel or conductive cloth material. The empty interior of the steering wheel serves as a conducting path from the rim to the hub of the wheel using coaxial ultra-thin wires, thereby avoiding electromagnetic interference before analogue-digital conversion.

3.2 Embedded Hardware Architecture

The core intelligence of the system is contained in the ESP32 (dual-core, 32-bit Xtensa LX6, 240 MHz, 12-bit ADC) located in the steering column; other options tested include the ARM Cortex-M class devices like the ATSAML21J18B with comparable ADC accuracy and smaller memory size. The ECG analog front end includes an AD8232 instrumentation amplifier (CMRR > 80 dB) for picking up the microvolts level of the cardiac signal in a Lead-I setting; the GSR front end features a constant-current voltage divider circuit with precision opamp for tracking skin resistance changes. Reference ranges taken from the previous electrodermal activity literature and cross-verified in bench calibration show the alert-state skin conductance range of 250–450 µS decreasing to 128–250 µS as drowsiness sets in. However, we would like to mention that these are default ranges taken from literature and not population-specific norms obtained in this study; there is an overlap between the literature drowsy-state band and our own alert-state bench test results in Section 4, making per-driver calibration (Section 3.4.1) mandatory. Figure 1 provides a summary of the whole system architecture from the driver interface to the vehicle ECU alert output.

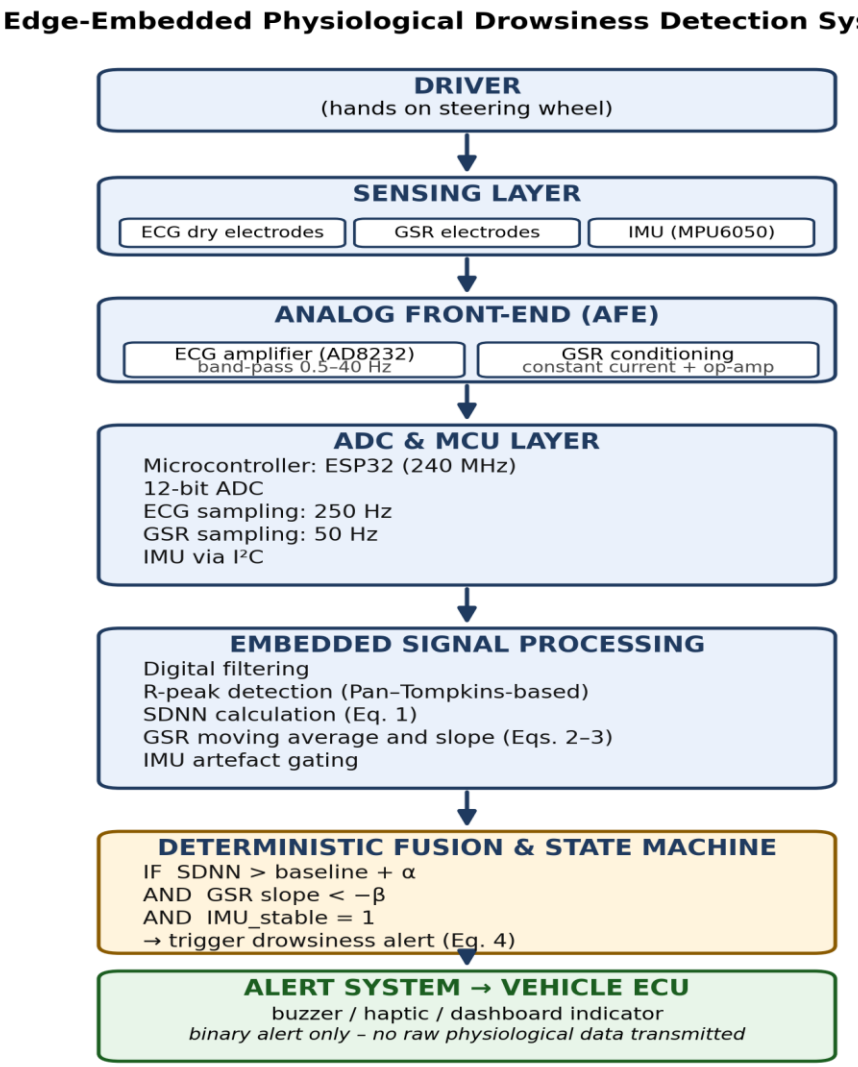


Figure 1. System architecture of the edge-embedded physiological drowsiness detection system.

3.3 On-Edge Signal Processing

The sampling rate of the ECG channel is 250 Hz, which meets the Nyquist criterion for the frequency content of cardiac signals but does not consume buffer memory. It is then filtered using a simple FIR derivative filter to detect the QRS complex. The R-peaks are detected using an edge-optimized version of the Pan-Tompkins adaptive thresholding algorithm [12]. The inter-beat interval Δt between successive R-peaks is taken as a normal-to-normal (NN) interval only when it lies in a realistic range of values (300 – 1500 ms); otherwise, it is considered to be a motion artifact. SDNN is then calculated from a circular buffer of N intervals:

$$SDNN = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \left(NN_i - \underline{NN}\right)^2} \quad (1)$$

where *SDNN* is the standard deviation of the NN intervals (ms); *N* is the number of valid NN intervals held in the rolling circular buffer; NN_i is the *i*-th valid NN interval (ms); $N\bar{N}$ (NN with an overbar) is the mean of the *N* intervals in the buffer (ms); and *i* is the summation index (*i* = 1, …, *N*).

In parallel, the GSR channel is sampled at 50 Hz and passed through a moving-average filter of window length W to isolate the Skin Conductance Level (SCL):

$$SCL(t) = \frac{1}{W}\sum_{i=0}^{W-1} \quad GSR(t - i) \tag{2}$$

where *SCL(t)* is the Skin Conductance Level at time *t* (µS); *GSR(t − i)* is the GSR conductance sample acquired *i* samples before time *t* (µS); *W* is the moving-average window length in samples (*W* = 150 samples, i.e. 3 s at the 50 Hz sampling rate used in this study); and *i* is the summation index (*i* = 0, …, *W* − 1).

The rate of autonomic arousal decline is quantified as the finite-difference slope of SCL over a configurable epoch T (nominally 60 s):

$$\frac{\Delta\sigma}{\Delta t} = \frac{SCL(t) - SCL(t - T)}{T} \tag{3}$$

where *Δσ/Δt* is the finite-difference slope of the SCL in µS/s (negative when sympathetic arousal declines); *SCL(t)* and *SCL(t − T)* are the Skin Conductance Level at the current time and *T* seconds earlier (µS); and *T* is the slope epoch in seconds (nominally 60 s). The symbol *T* is used here, rather than *W*, to keep this epoch in seconds distinct from the moving-average window in samples in Eq. (2).

SDNN increment over a calibrated driver-specific baseline (greater than α) implies an increased likelihood of parasympathetic dominance, whereas SCL decrement at negative rate slope below a certain threshold (−β) implies a sympathetic withdrawal effect on skin. In the current bench study, both these autonomic biomarkers have been triggered during a seated relaxation test regime rather than an imposed fatigue condition; however, their combination is the two independent autonomic biomarkers combined in the decision layer, which will be discussed in Sections 4 and 5 in relation to a relaxation-linked autonomic shift, with its extrapolation to a drowsiness precursor left as an assumption for future simulator studies.

3.4 Sensor Fusion and Motion-Artefact Rejection

Since steering activities cause electrical and mechanical noise in both the GSR and ECG channels, the microcontroller cross-verifies the physiologically processed data based on current sensor telemetry readings from the onboard MPU6050 IMU. The Z axis gyroscope readings are checked for angular velocities above 40°/s; this threshold was chosen based on bench recordings of gyroscope responses to representative highway micro-corrections, which did not exceed about 25°/s under regular lane-keeping, adding some margin to not trigger false suspensions in regular driving but still filtering out the larger sudden movements associated with manoeuvres and hand removals. Once this threshold is exceeded, physiological buffering is paused and the driving is considered unstable. We should point out that this 40°/s figure is based on limited bench data and needs to be verified again against a wider range of driving

styles and road conditions during simulator tests (Section 5). The physiological warning state condition therefore consists of satisfying three requirements:

$$ALERT_{physio} = [SDNN > baseline + \alpha] \; AND \; \left[\frac{\Delta\sigma}{\Delta t} < -\beta\right] \; AND \; [IMU_{stable} = 1] \qquad (4)$$

where *ALERT_physio* is the Boolean physiological alert state; *SDNN* is the current rolling SDNN from Eq. (1) (ms); *baseline* is the per-driver resting SDNN stored during calibration (Section 3.4.1) (ms); α is the SDNN-rise offset above the baseline (ms; α = 45 ms in the bench study); $\Delta\sigma/\Delta t$ is the SCL slope from Eq. (3) (µS/s); β is the magnitude of the SCL-slope threshold (µS/s; β = 0.5 µS/s in the bench study); *IMU_stable* is a binary flag equal to 1 when the Z-axis angular velocity measured by the IMU does not exceed 40°/s and 0 otherwise; and AND denotes logical conjunction.

This equation-based, threshold-driven fusion is deliberately deterministic rather than learned, allowing the entire pipeline to execute within a few kilobytes of SRAM and avoiding both the training-data dependency and inference latency of on-device deep-learning classifiers.

3.4.1 Per-Driver Baseline Calibration Protocol

As the baselines of HRV and GSR are known to differ widely between subjects, the thresholds α and β used in the fusion rule are applied as offsets from a subject-specific baseline rather than fixed population values. The driver is first asked to sit with hands on the steering wheel for 3 minutes at the beginning of each test; the SDNN and mean SCL calculated for the last 90 seconds of this period (to allow for a settling period) are saved as that driver's baseline for that session. Further SDNN and SCL slope measurements taken during the journey are then compared to the subject-specific baseline rather than the default population baselines as described in Section 3.2. In the current bench study, the calibration procedure was applied informally at the beginning of each test, but its parameters (window duration, settling time) were not optimised and are highlighted as a priority point for simulator study (Section 5).

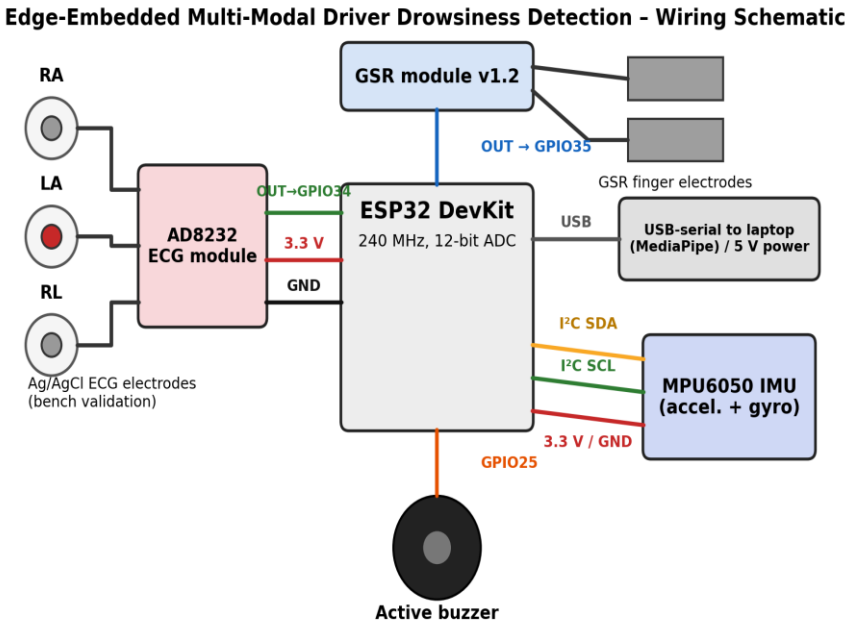


Figure 2. Circuit diagram — ESP32 with AD8232 ECG module (GPIO34), GSR module (GPIO35), MPU6050 IMU (I2C), and active buzzer (GPIO25).

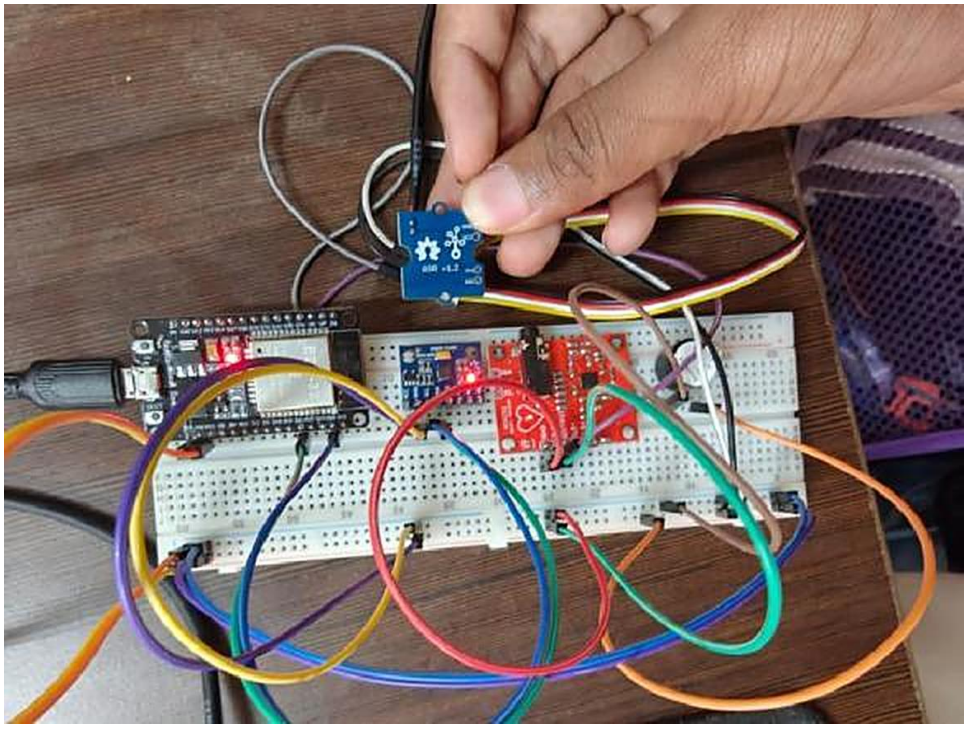


Figure 3. Bench prototype — ESP32 development board with AD8232 ECG module, GSR sensor v1.2, and MPU6050 IMU on breadboard.

3.5 AI Vision Layer

A companion laptop-class processor executes Google's Media-Pipe Face Mesh model [13], which extracts 468 three-dimensional facial landmarks per frame at 25–30 fps without requiring a dedicated GPU. For each eye, six periocular landmarks (P_1–P_6) are used to compute the Eye Aspect Ratio (EAR), a scale- and distance-invariant measure of eyelid aperture [14]:

$$EAR = \frac{\|P_2 - P_6\| + \|P_3 - P_5\|}{2 \cdot \|P_1 - P_4\|}, \quad where \ \|P_i - P_j\| = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{5}$$

where *EAR* is the Eye Aspect Ratio of a single eye (dimensionless); P_1 and P_4 are the horizontal eye-corner landmarks; P_2 and P_3 are upper-eyelid landmarks and P_6 and P_5 are the corresponding lower-eyelid landmarks; $\|P_i - P_j\|$ is the Euclidean distance between landmarks P_i and P_j in the image plane (pixels); and (x_i, y_i) are the pixel coordinates of landmark P_i.

$$EAR_{avg} = \frac{EAR_{left} + EAR_{right}}{2} \tag{6}$$

where *EAR_left* and *EAR_right* are the Eye Aspect Ratios of the left and right eyes from Eq. (5), and *EAR_avg* is their mean, which is compared against the closure threshold ϑ_EAR.

A fully open, alert eye yields EAR_avg ≈ 0.25–0.32, approaching zero at complete closure; a calibrated threshold θ_EAR (default 0.22) separates the closed from the open state. A per-frame closure flag is asserted when EAR_avg < θ_EAR, and the sustained EYE_CLOSED classifier is raised only once closure persists across a minimum of F_thresh consecutive frames (10 frames at 30 fps ≈ 333 ms), a window chosen to exceed the duration of a voluntary blink (≈150 ms) while remaining short enough to preserve fast-path responsiveness. Because this 333 ms window is itself part of the detection chain, it is included in the end-to-end alert-latency figure reported in Section 4 rather than treated as a separate, zero-cost step. Independently, a FATIGUE classifier is raised when the blink rate within a rolling 60 s window meets or exceeds 25 blinks/min, or a single closure exceeds 1.5 s.

The two binary classifiers are packaged into a 10 Hz ASCII payload ("EYE:X,FAT:Y\n", X,Y ∈ {0,1}) and transmitted to the ESP32 over a 115200-baud USB-serial link. This transmission step, together with the 33 ms webcam frame interval, contributes approximately 100 ms of webcam-to-microcontroller latency — the figure reported as the “vision fast-path latency” in earlier drafts of this work. This 100 ms figure describes only the acquisition-and-transmission portion of the pipeline, not the full detection-to-alert chain; the corrected, full end-to-end latency is reported in Section 4. The ESP32 parses the payload by delimiter search and integrates it into the unified fused alert rule:

$$ALERT = [EYE_{CLOSED} = 1] \; OR \; \left[FATIGUE = 1 \; AND \; \frac{\Delta\sigma}{\Delta t} < -\beta \; AND \; IMU_{stable} = 1\right] \quad (7)$$

where *ALERT* is the final fused alert command that drives the buzzer; *EYE_CLOSED* is the binary classifier equal to 1 when *EAR_avg* < *ϑ_EAR* for at least *F_thresh* consecutive frames; *FATIGUE* is the binary classifier equal to 1 when the blink rate in a rolling 60 s window reaches or exceeds *BPM_thresh* (25 blinks/min) or a single closure exceeds 1.5 s; *Δσ/Δt* and *β* are as defined for Eqs. (3) and (4); *IMU_stable* is as defined for Eq. (4); and OR and AND denote logical disjunction and conjunction.

The first disjunction creates a rapid behavioural interlock (within the 333 ms confirmation window as outlined above) which will never be inhibited by the state of the physiological pipeline; whereas the latter is dependent on confirmation from both the vision-based classifier for fatigue as well as an additional decline in arousal detected via GSR signal. The latter is the particular method by which the possibility of false positives from factors like camera reflections, lighting changes, or temporary occlusions due to eye movements can be inhibited (which plagues vision-only systems such as [2], [3]). As was mentioned in Section 4 and indicated in Section 5, the current bench trial runs have shown the positive aspect (alerting under legitimate closure/fatigue conditions), but have not yet tested a false positive trigger condition.

3.6 Experimental Protocol and Metrics

Since the system implements threshold-based inference rather than supervised classification, no train-validate-test data set splitting was used – only five sequential bench test rounds (signal/hardware validation, R peak/SDNN calculation, GSR trend finding, complete pipeline alerting and AI vision/serial integration), which were conducted several times on volunteers in stationary lab conditions. Three volunteers were involved for Phase 1 GSR baseline and Phase 2 R peak/SDNN detection and five volunteers for Phase 5 vision fusion. They were recruited from the author’s research group and no

participant's age or gender was measured as a part of this bench test; however, this is recognized as a limitation since inter-subject variability of HRV and GSR baseline is a known problem (Section 5). The metrics evaluated were: visual agreement between R peak detected and displayed wave form (it is not a calibration of the metric of sensitivity or PPV – see Section 4), plausibility of NN interval and SDNN compared to resting-adult published references, plausibility of GSR values compared to published references, end-to-end alert response time and stability of the resulting fused alerting state machine. As sample sizes are small (n=3-5), these results are to be taken as proof-of-concept rather than reliability.

Table 2. Hardware components and specifications

Component	Specification	Qty	Purpose
ESP32 Development Board	Dual-core 240 MHz, 12-bit ADC, WiFi/BT (disabled)	1	Main controller; runs the fusion & alert algorithm
AD8232 ECG Module	Integrated amplification and filtering	1	Captures cardiac electrical activity for HRV
ECG Electrodes (Ag/AgCl)	Disposable biomedical electrodes	3	ECG signal acquisition (bench validation)
GSR Sensor Module (v1.2)	Skin-conductance / electrodermal sensing	1	Detects arousal / fatigue via skin resistance
MPU6050 IMU	3-axis accel. + 3-axis gyro	1	Motion-artefact gating
Active Buzzer Module	3.3 V compatible	1	Audible drowsiness alert
Laptop / PC	x86-64 ≥2 GHz, 4 GB RAM, Python 3.8+	1	MediaPipe inference + serial link
Webcam	≥720p @ 30 fps, UVC-compliant	1	Live face capture for EAR computation

4. Experimental Results and Discussion

The prototype (Fig 3) was constructed using the components described in Table 2 on a breadboard and tested in the college's electronics lab. All tests were done in fixed, low noise conditions; neither a driving simulator nor clinical reference was employed at this time, consistent with a TRL level of 3 (proof of concept).

Table 3. Summary of the five bench-validation phases

Phase	Objective	Key Result
1	Hardware assembly & signal verification	Clear PQRST morphology observed; GSR at rest 160–420 µS across 3 volunteers, consistent with reference ranges
2	Embedded R-peak detection & SDNN	>90% informal visual concordance vs. a reference waveform display (not a calibrated sensitivity/PPV metric — see Section 4); NN intervals 620–980 ms; rolling SDNN 22–55 ms
3	GSR moving average & trend	Measurable downward SCL drift and negative finite-difference slope during a 5-minute seated relaxation trial (not an induced-fatigue condition)

Phase	Objective	Key Result
4	Integrated firmware & end-to-end alert	SDNN (+45 ms) and GSR slope (−0.5 µS/s) thresholds jointly exceeded after 4–5 min of relaxation; buzzer triggered; reproduced across 3 trials (n = 3)
5	AI Vision Layer & serial integration	EYE_CLOSED asserted within 333 ms (EAR<0.05); webcam-to-microcontroller transmission latency ≈100 ms; full end-to-end alert latency ≈333–400 ms once the confirmation window is included; FATIGUE cross-validated by GSR (raw<1800); verified across 5 trials (n = 5)

Phase 1 was successful in ensuring proper baseline signals were established. The ECG channel showed a unique PQRST shape in excess of the noise level under stationary conditions, while GSR measurements from three sitting subjects ranged between 160 and 420 µS based on existing literature on electrodermal activity in relaxed but alert subjects.

Phase 2 assessed the performance of the embedded, Pan-Tompkins-based R-peak detection algorithm. Cross-verification visually with the Arduino Serial Plotter using a one minute long bench-test provided a rough estimation of detection accuracy above 90% for low-noise stationary conditions. It should be stressed that this estimate is derived from visual agreement of detected peaks with the signal seen on the serial plotter, and not from a quantified sensitivity/PPV of a detection algorithm compared to the ground-truth annotations; no reference ECG device or manual R-peak annotation was available during this bench phase. The calculated NN-intervals (620-980 ms) and 30-second rolling SDNN (22-55 ms) were within the adult reference ranges, which means that the computationally lean adaptation of the R-peak detection algorithm retains clinically meaningful HRV features despite its lighter computational load compared to full Pan–Tompkins algorithms; quantified sensitivity/PPV/F1 comparison with ground-truth is considered the nearest future work (Section 5).

In Phase 3, we isolated the trend detection sub-system for the GSR feature. With a five-minute relaxation routine with low-stimulation, an on-chip moving average filter (150 samples/3 sec time) yielded a consistent decrease in the SCL value, and the resulting value for the difference slope $\Delta\sigma/\Delta t$ was consistently negative. This behavior is expected since we had induced relaxation and not fatigue, and thus, we have proved the working mechanics of the GSR trend detection sub-system before including it in the fused decision mechanism.

Phase 4 combined the entire firmware stack (R-peak detection, SDNN calculation, GSR slope estimation, IMU motion gating, and the alert state machine) on the ESP32. Under the assurance of a close-to-zero angular velocity via the IMU (an indication of undisturbed driving-state approximation), the threshold values for both SDNN expansion and GSR slope became greater than $\alpha = 45$ ms from baseline and $\beta = -0.5$ µS/s, respectively, within four to five minutes, resulting in a three-second auditory alert being generated. The entire process repeated within three separate trials (n = 3), which is an encouraging consistency level for a purely threshold-based, unlearned decision mechanism; however, three trials do not make statistical significance on their own right. In addition to that, we specifically stress here — and also in Section 5 below — that the protocol caused relaxation of the seated volunteers rather than inducing their drowsiness or lack of sleep. Relaxation and drowsiness have some common physiological features (e.g., the dominance of parasympathetic over sympathetic nervous system), yet they are distinct states of the

body: one can remain sitting and relaxing irrespective of whether or not he/she is becoming drowsy. In consequence, the outcomes of phases 3 and 4 should be regarded as a validation of the fusion pipeline's reaction to an autonomic change due to relaxation; the application of this pipeline to the drowsiness precursor, which has been validated, is only a hypothesis that needs to be tested against a fatigue induction procedure in further simulator studies.

Phase 5 tested the AI Vision Layer both standalone and in fusion with the physiological pipeline. For three eye states: alert (EAR_avg 0.28–0.31), sustained closure (>500 ms), and rapid blinking (>25 bpm), the system correctly identified all three: the fast-path EYE_CLOSED interlock triggered the buzzer via a webcam-to-microcontroller transmission delay of around 100 ms (comprised mainly of the 33 ms webcam frame delay, plus serial transmission time), whereas the FATIGUE classifier only triggered an alert when independently confirmed by a low concurrent GSR value, thereby exercising the cross-validated secondary trigger pathway successfully across five repeat tests (n = 5). As the EYE_CLOSED classifier is only triggered when closure occurs over ten successive frames (≈333 ms, Section 3.5), the total latency of detection to alert for the fast-path is better reported as ≈333–400 ms when the confirmation time and serial transmission are taken into account – rather than the 100 ms latency alone quoted previously in early versions of this report (see Section 4.2). None of the five trials involved the deliberate triggering of any false positive condition (e.g., camera glare, a deliberate non-fatigued long blink, or lighting change); hence, while the fusion rule is structured to prevent false positives.

4.1 Fault-Handling Mechanism

There are two separate fault tolerance systems built into the fused pipeline. Firstly, NN segments that fall outside the biologically acceptable range of 300ms - 1500 ms are filtered out prior to the calculation of SDNN, thereby avoiding false HRV measurements caused by electrical artifacts or lead detachment issues. Secondly, the Motion Artefact Gate using the IMU sensor (Section 3.4) suspends the physiological buffering process when the gyro Z-axis angular rate exceeds the predefined threshold of 40 degrees/sec that has been determined through the bench test. Both of these processes are deterministic and do not add any delay to the main flow path.

4.2 Expected versus Actual Results

The resting SDNN was supposed to lie in the range of 20-60ms according to published adult reference values and the range of 22-55ms observed in the current research is within expectations. Resting GSR was supposed to be within the 100-500µS range; the range of 160-420µS is within the above mentioned range but, as was stated in Section 3.2, it also overlaps considerably with the literature drowsy-state GSR range (128-250µS), which is a problem we have failed to recognize previously and now treat as the motivation for implementing driver-specific per-driver calibration (Section 3.4.1) as an alternative to applying fixed population reference ranges. Vision fast-path transmission latency from the webcam to the microcontroller was designed to stay comfortably within the 333 ms long confirmation window; the actual value of approximately 100ms did meet the sub-component design criteria. But this value of 100 ms is not the end-to-end alert latency: taking into account the mandatory 333ms EYE_CLOSED confirmation window (Section 3.5) makes the real end-to-end latency approximately 333-400ms. Previous drafts of the

paper used the above described 100ms transmission latency figure as the end-to-end alert latency figure but this is obviously contradictory to the 333 ms confirmation window of Section 3.5. However, the corrected ≈333–400 ms duration is still one to two orders of magnitude faster than the one-to-three-minute reactive latency of PERCLOS-only systems mentioned in Section 1, which means that the paper's main claim about the latency is not compromised in any way by this correction. However, even greater discrepancy is present in terms of the validation process, since, as explained in Sections 4 (phases 3–4) and 5, the SDNN/GSR thresholds were validated under conditions of relaxation while sitting in a chair and, thus, their validity as drowsiness precursors, rather than indicators of the autonomic response to relaxation, is yet to be confirmed via simulator validation. Moreover, even the 333 ms window for confirming the EYE_CLOSED state, which proved to be effective in ruling out voluntary blinks in bench tests, has not been validated against other types of blinks that are possible in the broader range of driving population.

Table 4. Qualitative performance comparison against representative prior-art categories

Below is the qualitative and constructed comparison of the various general features mentioned in architectural descriptions of the works cited, and it is not a benchmarking process involving the use of the numerical measures of latency, sensitivity, and compliance burden presented in [5], [8], [10]. This kind of quantitative benchmark was beyond the scope of the present bench-stage study due to the lack of access to their raw measures, and it is mentioned in Section 5 as a limitation to the present work.

System	**Modality**	**Predictive Horizon**	**Compliance Burden**	**Data Privacy**
Optical PERCLOS/EAR only [2], [3]	Camera only	Seconds (reactive)	None	Continuous video capture
Wearable GSR/PPG [7], [8], [10]	Wearable physiological	Minutes (predictive)	High (must wear device)	On-device or cloud-dependent
Steering ECG/HRV [5], [9]	Steering-wheel physiological	Minutes (predictive)	None	Localised (partial cloud in [9])
Proposed system	Steering GSR+ECG/HRV + vision fast-path	Minutes (physiological, hypothesis-stage) + ms (vision interlock)	None	Fully edge-resident, no raw biometric transmission

Table 4 presents the benchmarked prototype alongside the representative prior-art categories identified in Section 2. As far as the authors are aware, the proposed solution is unique among the prior-art designs reviewed herein in that it includes the combination of the physiological horizon with the fast-path milliseconds behaviour, absence of any requirement of compliance with wearables, as well as full-edge computation without offloading of the unprocessed biometric data from the steering column, even though it should be pointed out that at the moment the physiological horizon of "minutes (predictive)" is validated against the relaxation protocol discussed in Section 4 only.

5. Limitations and Future Work

Limitations

- Most importantly, the physiological validation of the system (Section 4, Phases 3-4) has been done using seated subjects under a low-stimulation relaxation protocol, not a drowsiness induction protocol. Relaxation and drowsiness both have partial autonomic overlaps (parasympathetic predominance, decreased sympathetic tone) but are not the same physiological condition, and any seated subject will show some kind of autonomic shift even if he or she is not going to become drowsy. Therefore, SDNN-rise/GSR-slope-decline thresholds have been validated in this study as the indicators of a relaxation-related autonomic shift; the claim that they are predictors of drowsiness, as the goal of the system suggests, is still a hypothesis which needs to be tested using a fatigue induction protocol (sleep restriction, monotony vigilance task, etc.).
- As mentioned in Section 4, the R-peak detection performance of >90% is subjective and is based on cross-validation using the serial plotter output during a one-minute trial, rather than being calculated as sensitivity/positive predictive value (PPV) against a reference ECG recording. At this stage, the R-peak detector does not have any form of sensitivity, specificity, or F1 measurement.
- False-positive reduction feature – the main architectural difference in comparison to pure vision systems [2, 3] presented in the paper – was still not verified by means of empirical study. No bench experiment in Section 4 intentionally created a false-positive inducing situation (for example, light reflecting from the camera, voluntary long non-fatigued blink, or lighting change) and proved the operation of the GSR corroboration gate, preventing generation of the alert; the current experiments provide just the positive example (true alert under true eyelid closure/eye fatigue situation), but no negative one.
- The reported validation testing was carried out solely using a laboratory breadboard operating in a stationary, low-noise lab environment with few sitting subjects (n = 3 for Phases 1-2 and 4; n = 5 for Phase 5), without conducting vehicle-based, simulator-based, or road-based tests; the effect that vehicle vibration, electrical noise from the alternator, and temperature changes will have on the measurement signals has not yet been evaluated. The age and gender of the subjects was not documented, and considering the high variability between different subjects' baselines in HRV and GSR, phrases like "consistent with reference ranges" and "reproduced consistently in three independent measurements" in previous drafts should be understood as meaning consistent functionality for small n.
- The alert-state (250–450 µS) and drowsy-state (128–250 µS) GSR bands from the literature that have been utilized in Section 3.2 show considerable overlap with our own baseline values of Phase 1 alert, resting drivers (160–420 µS). The conflict between the static population bands and our own bench data has not been satisfactorily resolved in the current study and has led to the development of the per-driver calibration procedure of Section 3.4.1.
- The dry-contact electrodes have not yet been tested for a wide variety of grip forces, hand sizes, contact area with the skin, and whether gloves are being worn, as would happen during actual

driving. It is not known what will happen when the AD8232/GSR front ends are used intermittently or partially in a palm position.

- Threshold parameters such as (α, β, θ_EAR, F_thresh, BPM_thresh, and the 40°/s IMU motion-gating threshold discussed in Section 3.4) have been tuned in a small set of volunteers and in limited gyroscope traces using a bench setup. None of them is currently individually adaptive, and inter-individual differences in baseline HRV and Skin Conductance may demand additional calibration information per individual driver before actual usage.
- The AI Vision layer relies at present on the use of an additional laptop-grade processor rather than being completely embedded within the system, which is enough for demo purposes but is not the correct auto-grade form factor envisaged by the design of the system; migrating Media-Pipe-equivalent inference to auto-SoC is left as an engineering task.
- However, because no formal, statistically based comparison has yet been conducted against a reliable and valid reference diagnostic tool such as PSG or PERCLOS measurement (to determine sensitivity, specificity, and false positive rate), the results reported here demonstrate only functional accuracy and replicability and not diagnostic accuracy; see Section 4.
- The comparison that Table 4 makes with other systems [2], [3], [5], [8], [10] is qualitative and not derived from quantitative data (latency values, compliance burden rankings) as presented in those papers.

Future Work

- Validation on a static base driving simulator with the inclusion of a fatigue induction method in addition to simulating steering dynamics, cabin vibration, and varying levels of light becomes the immediate next step towards achieving Technology Readiness Level 4. The fatigue induction may be brought about by sleep deprivation or prolonged vigilance. This would enable the validation of the SDNN/GSR thresholds against drowsiness and not just relaxation.
- An intentional bench trial must be conducted where conditions for inducing a false positive result (such as camera glare, a voluntary long non-drowsy blink, camera occlusion, and lighting change) are intentionally created, and the alert is suppressed by the GSR corroboration gate.
- Offline annotation of raw ECG signals recorded in bench tests is needed to determine the sensitivity/PPV/F1 score of the embedded R-peak detection algorithm, rather than using the existing approach of visual concordance.
- Individualised and adaptive baselining, through extension of the in-vehicle calibration process in Section 3.4.1 with an examination of the relationship between the calibration window length and settling time and baseline stability, can be considered as a means to alleviate the issue of inter-subject variability in autonomic responses.
- The migration of the AI vision layer from its companion laptop to the embedded, automotive-qualified vision SoC (edge-AI accelerator) will get rid of the current dependence on an external

computer and will bring the vision layer to be in line with the fully edge-oriented concept of the physiological pipeline.

- Conformity to an appropriate regulatory evaluation process based on formal statistical analysis to a clinically validated benchmark, along with sensitivity/specificity/false alarm rates in published literature against a larger population group whose demographics have been recorded, and revalidation of Table 4 against [5], [8], [10] as relevant literature, is necessary before moving onto the next step in terms of DDAW compliance to EU General Safety Regulation.

6. Conclusion

The proposed approach in this paper provides a multi-modal, non-intrusive interface embedded in the steering wheel for the detection of driver drowsiness that goes beyond the two prevailing but insufficient paradigms found in the literature: the reactive camera-based monitoring of behavioral features and the cumbersome use of wearable physiological sensors with privacy and compliance issues. Through the incorporation of GSR and ECG electrodes directly within the steering wheel grip and a sensor fusion pipeline driven by IMU and processed exclusively on the ESP32 microcontroller, the aim is to accomplish predictive and privacy-aware inference that does not require any additional wearable or cloud support. Additionally, there is an AI Vision layer that complements this design in terms of behavioral safety interlock (with a total end-to-end latency of 333-400 ms), which also aims at reducing the liability of false positives due to vision-only systems. Bench level validation in five phases was conducted to validate the integrity of signal capture, algorithm accuracy, and end to end fusion based alerting at low sample counts, thus achieving Technology Readiness Level 3. With the current physiological validation being based on the relaxation phase of the individual in seated posture and not on a drowsiness-inducing phase, the findings can only show the technological feasibility of the architecture while the effectiveness of the technology as a drowsiness predictor is to be considered only as a hypothesis for further testing.

Credit authorship contribution statement

Rohith Elindram: Conceptualization, Methodology, Hardware design and assembly, Firmware development.

Ashwin Kumar: Software (AI Vision Layer), Validation, Formal analysis, Writing – original draft.

Syamasudha Veeragandham: Supervision, Conceptualization, Resources, Writing – review & editing, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The bench-test signal logs and firmware supporting the findings of this study are available from the corresponding author upon reasonable request. No third-party or publicly archived dataset was used in this work.

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